Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
STEPS TO SOLVE	1. Cross multiply ($\frac{a}{b}=\frac{c}{d} \rightarrow a\cdot d = b\cdot c$) 2. Distribute (if necessary) 3. Solve!

GUIDED EXAMPLES

1. $\frac{4}{3}=\frac{x}{6}$

$3\cdot x = 4\cdot 6$

$\frac{3x}{3}=\frac{24}{3}$

$\boxed{x=8}$

2. $\frac{2}{10}=\frac{4}{n+5}$

$2(n+5)=10\cdot 4$

$2n+10=40$

$-10 \quad -10$

$\frac{2n}{2}=\frac{30}{2}$

$\boxed{n=15}$

3. $\frac{y-3}{2}=\frac{y}{5}$

$2y=5(y-3)$

$2y=5y-15$

$-5y \quad -5y$

$\frac{-3y}{-3}=\frac{-15}{-3}$

$\boxed{y=5}$

4. $\frac{2k-3}{k+9}=\frac{3}{5}$

$5(2k-3)=3(k+9)$

$10k-15=3k+27$

$-3k \quad -3k$

$7k-15=27$

$+15 \quad +15$

$\frac{7k}{7}=\frac{42}{7}$

$\boxed{k=6}$

YOU TRY!

Directions: Solve each proportion.

1. $\frac{x+10}{8}=\frac{x}{3}$

$8x=3(x+10)$

$8x=3x+30$

$-3x \quad -3x$

$\frac{5x}{5}=\frac{30}{5}$

$\boxed{x=6}$

2. $\frac{p+2}{p+7}=\frac{3}{6}$

$3(p+7)=6(p+2)$

$3p+21=6p+12$

$-3p \quad -3p$

$21=3p+12$

$-12 \quad -12$

$\frac{9}{3}=\frac{3p}{3}$

$\boxed{p=3}$

3. $\frac{w-2}{w}=\frac{3}{6}$

$3w=6(w-2)$

$3w=6w-12$

$-6w \quad -6w$

$\frac{-3w}{-3}=\frac{-12}{-3}$

$\boxed{w=4}$

4. $\frac{n-4}{6}=\frac{n-7}{9}$

$6(n-7)=9(n-4)$

$6n-42=9n-36$

$-6n \quad -6n$

$-42=3n-36$

$+36 \quad +36$

$\frac{-6}{3}=\frac{3n}{3}$

$\boxed{n=-2}$

5. $\frac{3}{2}=\frac{a-3}{a+1}$

$3(a+1)=2(a-3)$

$3a+3=2a-6$

$-2a \quad -2a$

$a+3=-6$

$-3 \quad -3$

$\boxed{a=-9}$

6. $\frac{25}{f+2}=\frac{5}{3}$

$5(f+2)=75$

$5f+10=75$

$-10 \quad -10$

$\frac{5f}{5}=\frac{65}{5}$

$\boxed{f=13}$

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7. $\frac{8}{m-9}=\frac{6}{m-8}$

$6(m-9)=8(m-8)$

$6m-54=8m-64$

$-6m \quad -6m$

$-54=2m-64$

$+64 \quad +64$

$\frac{10}{2}=\frac{2m}{2}$

$m=5$

8. $\frac{2}{r+4}=\frac{4}{r}$

$2r=4(r+4)$

$2r=4r+16$

$-4r \quad -4r$

$\frac{-2r}{-2}=\frac{16}{-2}$

$r=-8$

9. $\frac{5}{2}=\frac{v-8}{v-2}$

$5(v-2)=2(v-8)$

$5v-10=2v-16$

$-2v \quad -2v$

$3v-10=-16$

$+10 \quad +10$

$\frac{3v}{3}=\frac{-6}{3}$

$v=-2$

10. $\frac{6}{k-1}=\frac{8}{k-6}$

$6(k-6)=8(k-1)$

$6k-36=8k-8$

$-6k \quad -6k$

$-36=2k-8$

$+8 \quad +8$

$\frac{-28}{2}=\frac{2k}{2}$

$k=-14$

11. $\frac{y-7}{y-9}=\frac{3}{4}$

$4(y-7)=3(y-9)$

$4y-28=3y-27$

$-3y \quad -3y$

$y-28=-27$

$+28 \quad +28$

$y=1$

12. $\frac{h-6}{9}=\frac{h+9}{4}$

$4(h-6)=9(h+9)$

$4h-24=9h+81$

$-4h \quad -4h$

$-24=5h+81$

$-81 \quad -81$

$\frac{-105}{5}=\frac{5h}{5}$

$h=-21$

13. $\frac{r}{2}=\frac{r+10}{4}$

$4r=2(r+10)$

$4r=2r+20$

$-2r \quad -2r$

$\frac{2r}{2}=\frac{20}{2}$

$r=10$

14. $\frac{7}{2x-8}=\frac{3}{x-6}$

$7(x-6)=3(2x-8)$

$7x-42=6x-24$

$-6x \quad -6x$

$x-42=-24$

$+42 \quad +42$

$x=18$

15. $\frac{2}{3u+7}=\frac{3}{4u+9}$

$2(4u+9)=3(3u+7)$

$8u+18=9u+21$

$-8u \quad -8u$

$18=u+21$

$-21 \quad -21$

$-3=u$

16. $\frac{2x-1}{5}=\frac{3x+1}{10}$

$10(2x-1)=5(3x+1)$

$20x-10=15x+5$

$-15x \quad -15x$

$5x-10=5$

$+10 \quad +10$

$\frac{5x}{5}=\frac{15}{5}$

$x=3$

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